



Project Proposal

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Solution

Team



SHOPMIND



AI POWERED E COMMERCE
PLATFORM



Team Name: AgentX

Team ID: 68

Institution: GLS

university





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THE PROBLEM



Modern e-commerce operations are deeply fragmented – inventory, orders, customer support, pricing, marketing, and logistics all managed in silos. Manual processes lead to delayed decisions, stock-outs, poor customer experiences, and significant revenue loss. Without intelligent automation, businesses cannot scale efficiently or respond to market changes in real time.





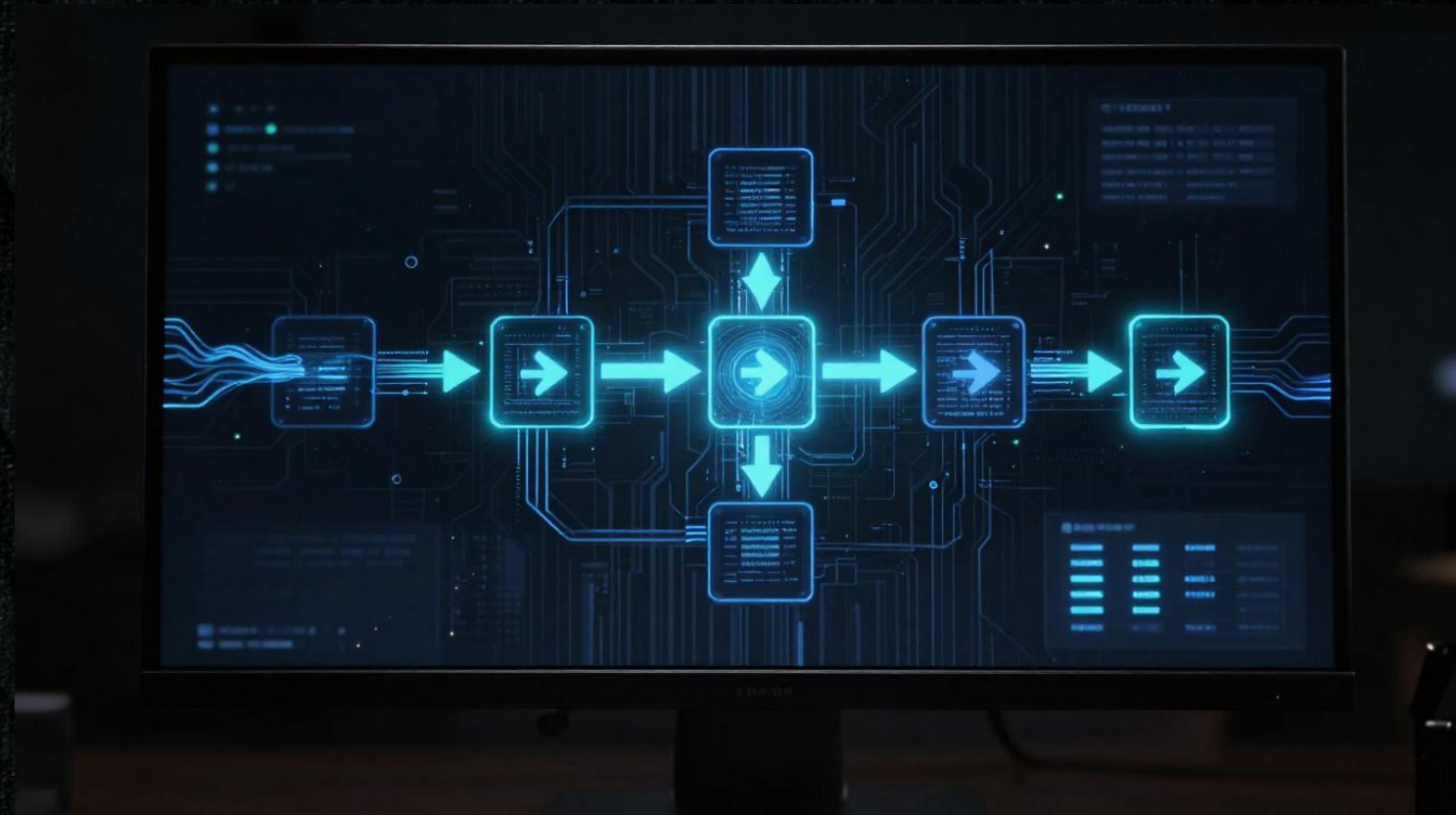
ShopMind is an AI-powered multi-agent platform that automates end-to-end e-commerce operations. By deploying specialised agents for inventory, orders, pricing, support and logistics, the system eliminates manual bottlenecks and enables real-time, data-driven decisions – while keeping humans in control of every high-risk action.

OUR SOLUTION



Autonomous agents collaborate through a shared orchestrator, continuously observing, reasoning and acting on live data. A built-in verification and feedback loop ensures accuracy, compliance and full auditability across all operations.





WORKFLOW DIAGRAM



PIPELINE FLOW

Data Sources → Event/API Layer → AI Agent Orchestration → Action Engine → Verification & Feedback Loop. Each stage passes structured context to the next, enabling real-time automated decision-making across all e-commerce operations.

AGENT LOOP

Observe → Reason → Act → Verify → Learn. Agents continuously monitor signals, reason over retrieved context, execute low-risk actions autonomously, verify outcomes, and update their knowledge base for improved future decisions.



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Inventory Agent

Stock Monitor & Auto-Reorder



Order Agent

Processing & Fulfillment

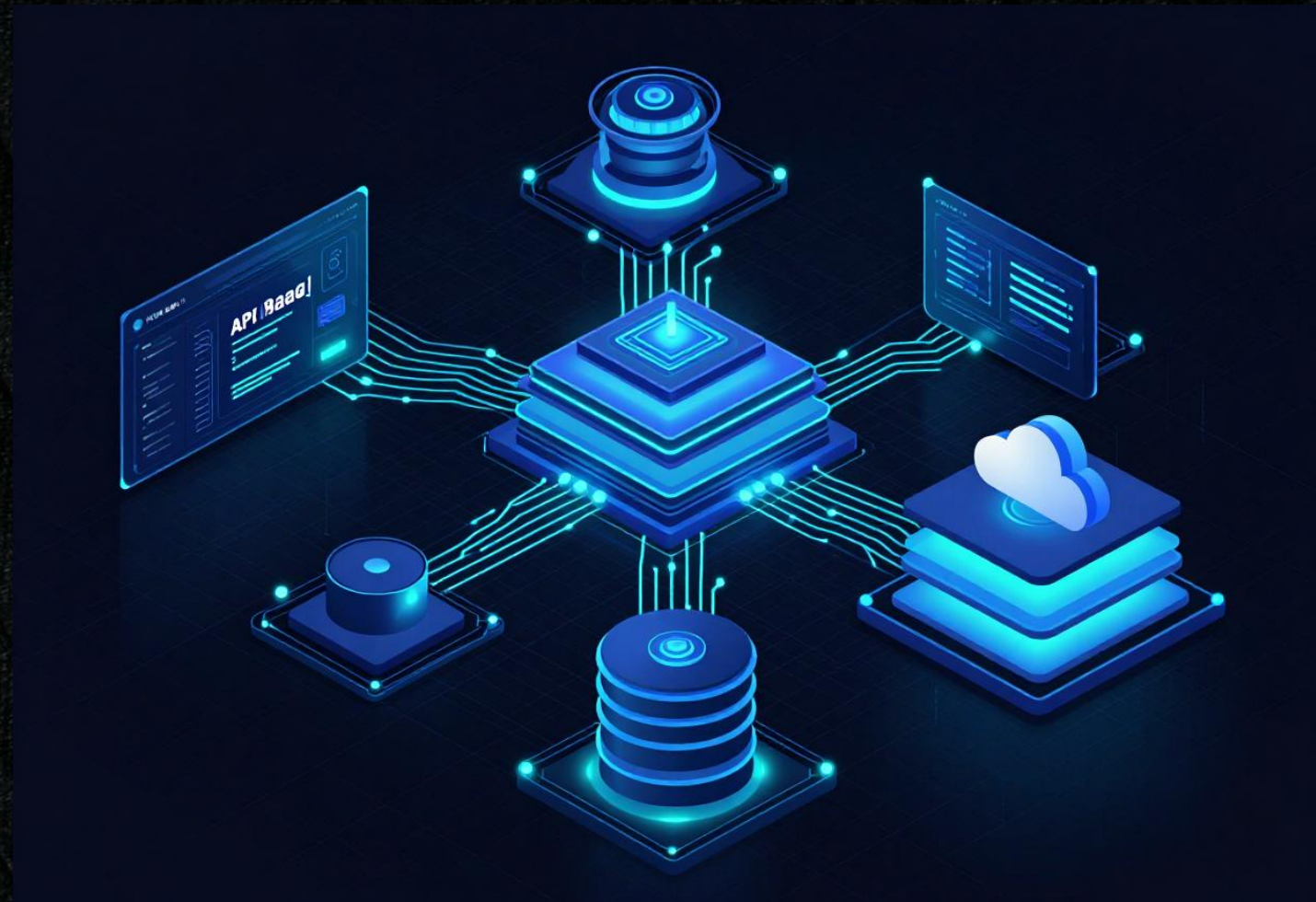


Support + Pricing Agent

CX, Pricing & Marketing AI



MULTI AGENT ECOSYSTEM



ARCHITECTURE



Frontend & Core Services

React/Next.js web dashboard, Python/FastAPI backend, JWT authentication with RBAC, AI agent orchestrator layer, LLM/ML inference engine, and RAG-powered knowledge base for contextual reasoning.



Infrastructure & Data Layer

PostgreSQL primary database, Redis queue for async tasks, e-commerce & logistics API integrations, AWS cloud deployment, Prometheus/Grafana monitoring, and secrets management via AWS Secrets Manager.





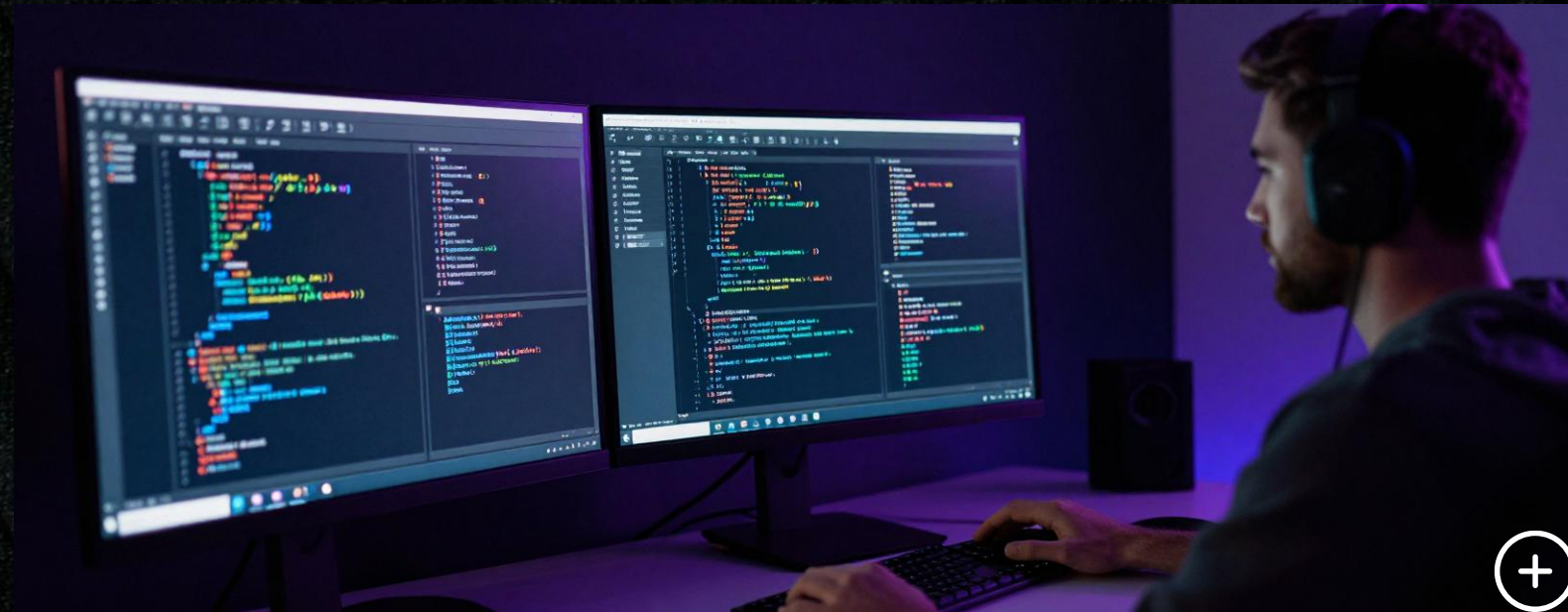
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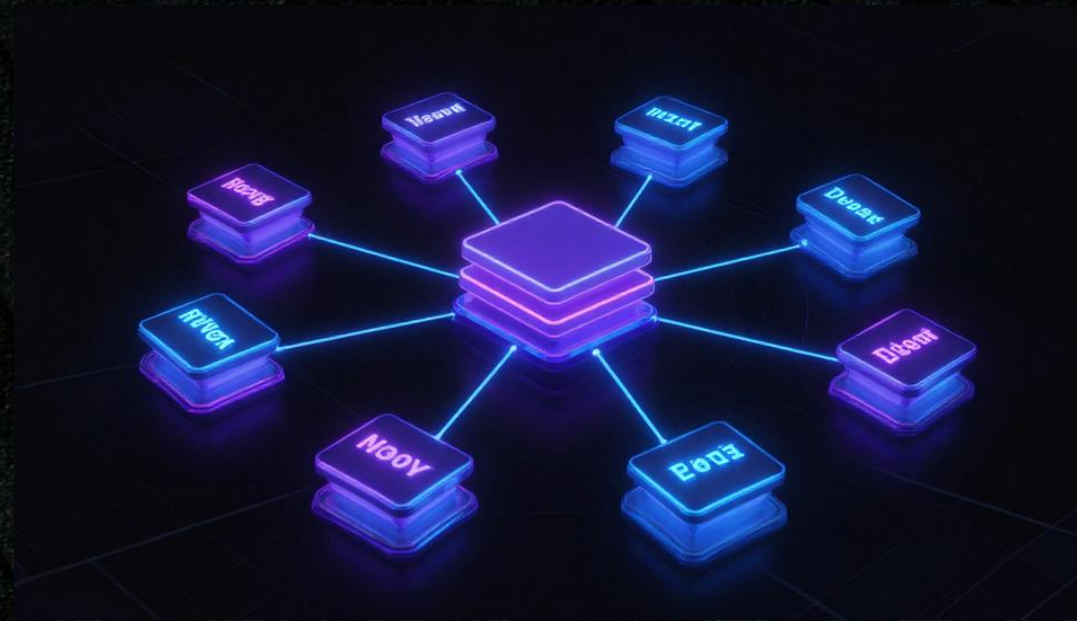


TECH STACK



Frontend: React / Next.js – modern, responsive dashboard UI
Backend: Python / FastAPI – high-performance async API layer
AI/ML: LLM + LangGraph – agent orchestration & reasoning
Database: PostgreSQL – relational data & audit logs
Cache/Queue: Redis – real-time event streaming & queuing
DevOps: Docker, Git – containerised CI/CD pipelines
Cloud: AWS ★ Preferred for final deployment
Data/Analytics: Python / Pandas – reporting & ML preprocessing





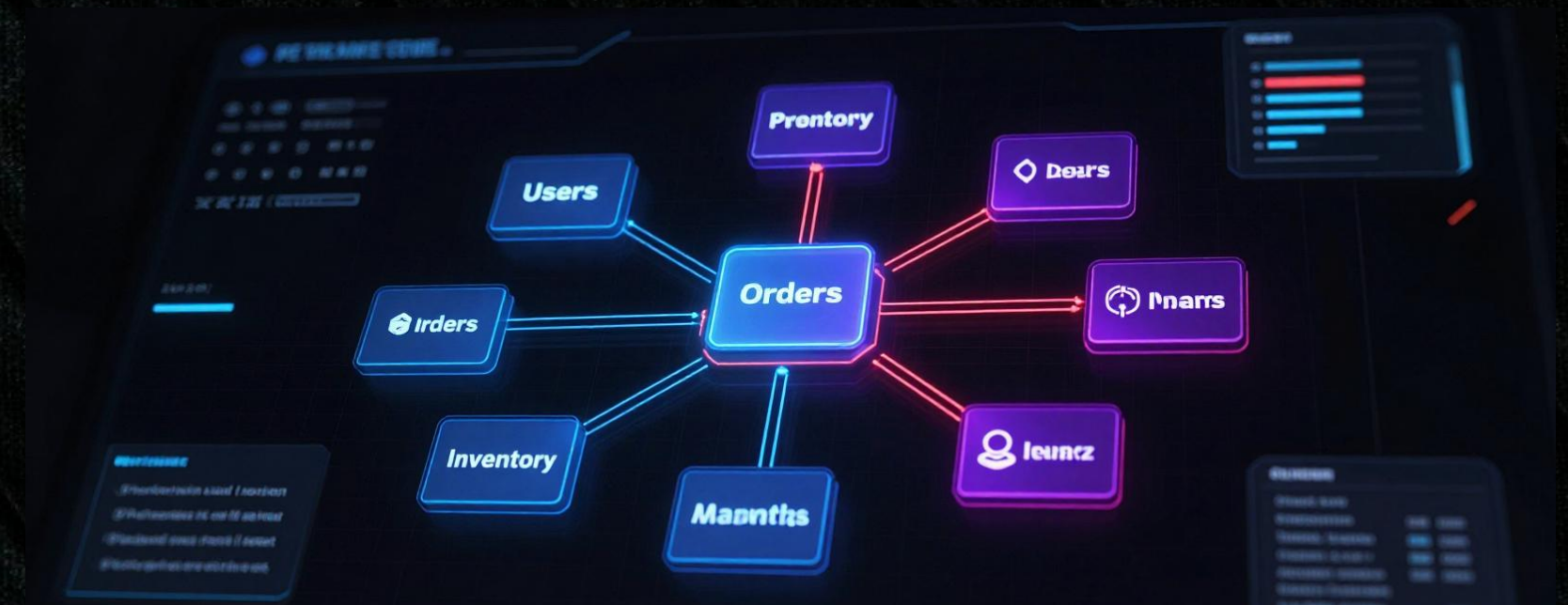
Core entities drive the ShopMind platform. Users link to Orders and Agent Memory via role-based access. Products connect to Inventory and Order Items, enabling real-time stock tracking. Orders aggregate Order Items and feed into Analytics. Agent Memory stores per-agent context for reasoning continuity. The Audit Log captures every agent action for traceability and compliance. Analytics aggregates across all entities for dashboard insights.



Key Entities:

- Users → Orders, Agent Memory
- Products → Inventory, Order Items
- Orders → Order Items, Analytics
- Agent Memory → Audit Log
- Audit Log → Analytics

DATA MODEL⁺





Event Ingestion → Context Retrieval → Agent Routing → Analysis → Risk Threshold Check. If risk is HIGH: trigger Human Approval gate – await operator decision before proceeding. If APPROVED or LOW-RISK: execute action via Action Engine. Post-execution: run Verification step to confirm outcome integrity.



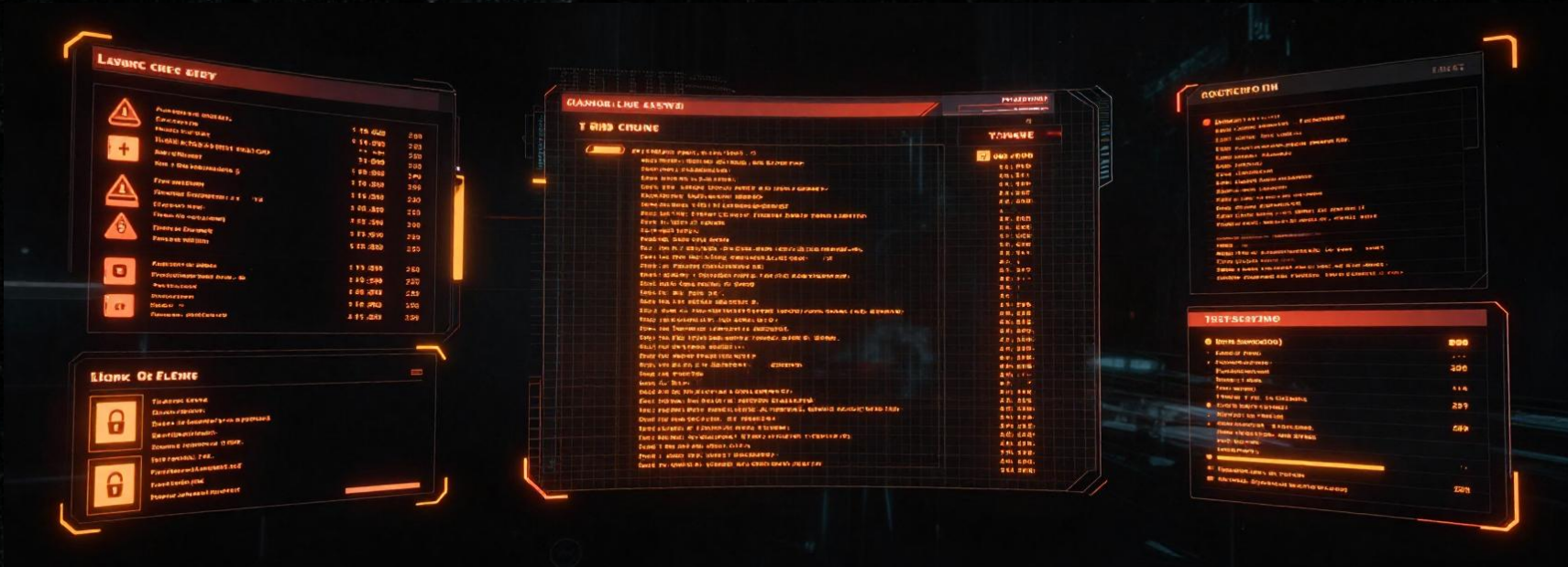
Audit Log records every decision, action, and outcome. Metrics are updated in real time. On failure or anomaly: initiate Retry logic (max 3 attempts), then escalate to human operator. Loop feeds back into Agent Memory for continuous learning and policy refinement.



PSEUDOCODE



ASSUMPTIONS, CHALLENGES & QUERIES



ASSUMPTIONS & RISKS: API / mock API availability and data access constraints; AWS resource provisioning and cost limits; reliance on synthetic vs. real e-commerce data. TECHNICAL CHALLENGES: LLM hallucination control in autonomous agent decisions; integration complexity across multi-vendor APIs and logistics; data privacy, security and RBAC enforcement; observability and traceability of agent actions at scale.

OPEN QUERIES: Which e-commerce APIs will be available or mocked for development? What is the confirmed AWS scope and budget ceiling? Will the system operate on real transaction data or synthetic datasets? What is the agreed autonomy level – fully automated execution or human-in-the-loop approval for all high-risk agent actions?



Phase 1 – Foundation

Project setup, DB schema, auth/RBAC, base API, CI/CD pipeline.

Phase 2 – Agents

Inventory, Order, Support & Pricing agents with LangGraph orchestration.

Phase 3 – Integration

E-commerce & logistics API connectors, Redis queues, event/API layer.

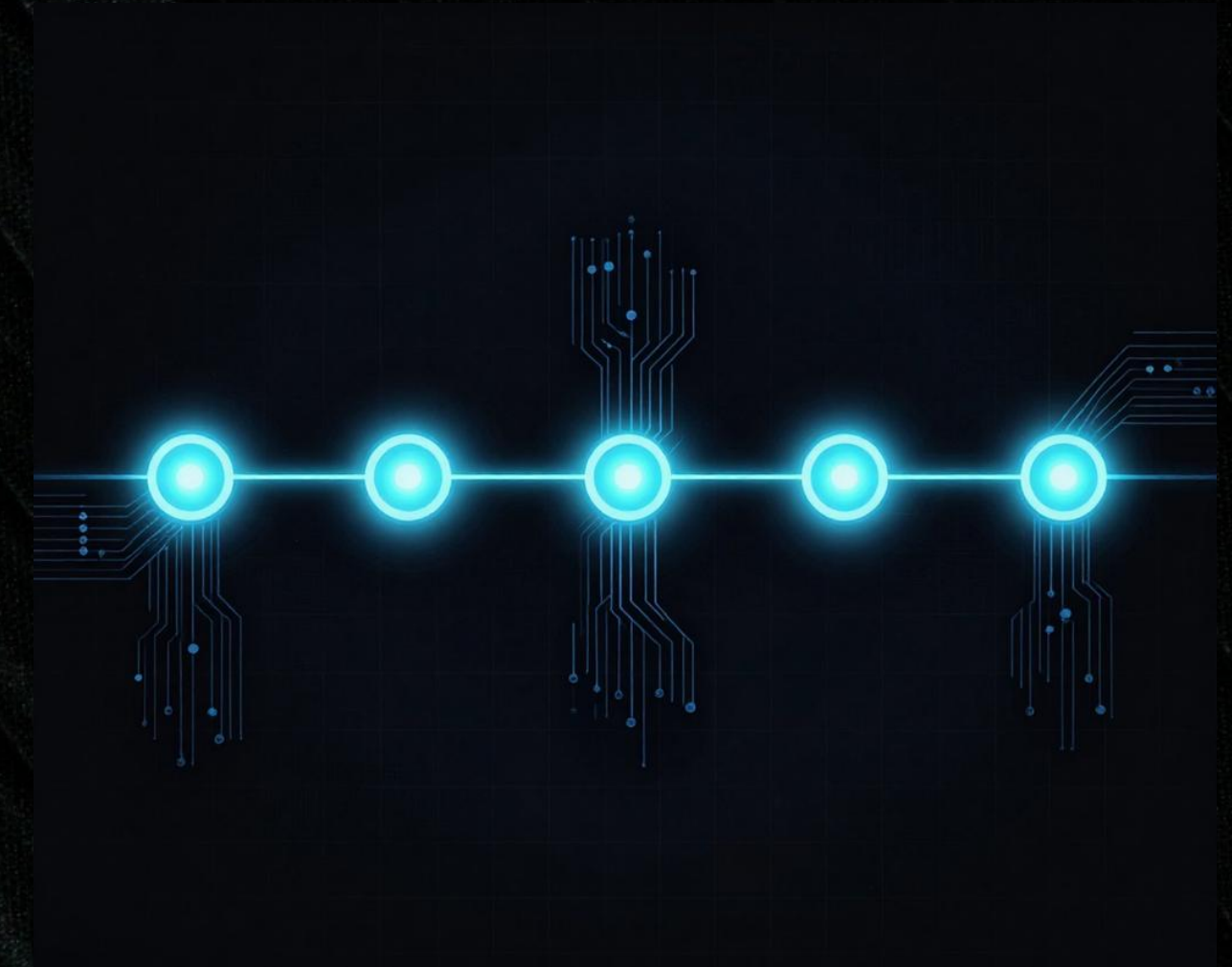
Phase 4 – Intelligence & Analytics

RAG knowledge base, ML models, real-time dashboard & audit logging.

Phase 5 – AWS Cloud Demo

Docker deployment on AWS, monitoring, secrets management & live demo.

IMPLEMENTATION ROADMAP



Success Criteria: All agents operational with <2s response time; human-in-the-loop approval flow functional; 95%+ audit log coverage; full AWS cloud deployment with live e-commerce demo environment.



QUESTIONS?

ShopMind – AI-Powered E-
Commerce

Automate.
Optimize. Scale.

21 August 2026 | Team
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